

# ”KNOWLEDGE GRAPH OF MONGOLIAN CINEMA”

## Team-1

| Role      | Name          |
|-----------|---------------|
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## Revision History

| Version | Date       | Description                                              | Author(s)                      | Reviewer(s) |
|---------|------------|----------------------------------------------------------|--------------------------------|-------------|
| v1.1    | 2025-11-18 | Initial Phase 1 draft: purpose, personas, scenarios, CQs | Sainbuyan L.,<br>Munkhnaran B. | Altangerel  |
| v1.2    | 2025-11-25 | Finalized Phase 1 for submission                         | Sainbuyan L.,<br>Munkhnaran B. | Altangerel  |
| v2.1    | 2025-12-03 | Phase 2: Information gathering, methodology              | Munkhnaran B.,<br>Altangerel   | Sainbuyan   |
| v2.2    | 2025-12-03 | Finalized Phase 2 for submission                         | Munkhnaran B.,<br>Altangerel   | Sainbuyan   |
| v3.1    | 2025-12-10 | Phase 3: Language Definition and ontology design         | Munkhnaran B.,<br>Sainbuyan L. | Altangerel  |

## Contributions

- Sainbuyan L.: Informal purpose, personas, scenarios drafting.
- Munkhnaran B.: Concept identification and competency questions.
- Altangerel N.: Relationships, ER conceptual model and focus classification.
- Sainbuyan L. (Project Manager): Consolidation, editing and submission on behalf of the team.

## PHASE 1-PURPOSE DEFINITION

### 1. Informal Purpose

When searching for information about Hollywood or Korean films, K-drama online, users can easily access comprehensive information in one place—release year, director, production company, genre, rating, viewer reviews, cast, and their related works. This is possible because these industries have established interconnected structured, knowledge ecosystems supported by mature knowledge graphs.

In contrast, the Mongolian film domain currently lacks such a unified, linked data ecosystem. Information about Mongolian cinema is scattered across many sources, unstructured, inconsistent and disconnected, difficult to integrate or analyze in a single search. Therefore, this project aims to:

- consolidate dispersed and unstructured Mongolian cinema data,
- build a connected data model using ontology and knowledge-graph methodology,
- provide a comprehensive, searchable, analyzable Knowledge Graph of Mongolian Cinema.

This knowledge graph will enable researchers, film enthusiasts, and media organizations to:

- explore connections among Mongolian films,
  - analyze relationships between filmmakers, actors, and studios,
  - study genres, award trends, and historical developments,
  - access all essential information about a film from a single query,
  - perform recommendations and graph-based analytics.
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### 2. Personas

#### Persona 1: Non-Fan Viewer

**Goal:** Has no film knowledge. Wants to pick a high-rated film by a director/actor his girlfriend likes for a movie date.

**Needs:**

- View linked information among director, film, actors
- Runtime and short summary
- High-rated films featuring a preferred artist

#### Persona 2: Film Enthusiast

**Goal:** Find other works by a favorite actor or director.

**Needs:** Similar-genre films, related works by the same creators, ability to discover underrated, lesser-known but high-rated films. Importance placed on rating and release year.

#### Persona 3: Business Owner

**Goal:** Collaborate with filmmakers popular among target customers (e.g., children, women).

**Needs:** Networks of artist collaborations, film ratings and recent releases, identify current trending filmmakers with strong portfolios.

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### 3. Scenarios

#### Scenario A: Non-Fan Viewer

Selecting a good Mongolian film for a date.

- **KG Needs:** Actor - list of other high-rated films; Movie - synopsis, runtime; Director – genre connections to understand film type.

#### Scenario B: Film Enthusiast

Finding underrated films by a favorite director.

- **KG Needs:** Director B – list of all directed films; Filter by high rating, low review count; Release year is before 2000.

#### Scenario C: Business owner

Collaborate with currently active, trending filmmakers.

- **KG Needs:** Production company - list of produced films; Filter by 2020–2024 high-rated releases; Identify frequently collaborating actors and directors.
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### 4. Competency Questions (CQs)

The knowledge graph must be able to answer:

1. Can Mongolian films be listed by genre?
  2. What films has Director X made, and in which years?
  3. Which actors starred in Movie Y?
  4. Which Mongolian actors have the highest number of film appearances?
  5. Which Mongolian films have won domestic or international awards?
  6. What works has Production Company Z released?
  7. What common features link films of a specific genre?
  8. Can we generate an actor-actor or actor–director collaboration network?
  9. Who are the trending filmmakers of the last five years?
  10. Can we visualize the career timeline of an actor or director?
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## 5. Concept Identification

**Table 1:** Concept Definition

| Concept            | Description                                                                            |
|--------------------|----------------------------------------------------------------------------------------|
| Movie              | Metadata including release year, runtime, genre, rating, summary, awards, review co    |
| Person             | General category for creative individuals (actors, directors, producers, cinematograph |
| Actor              | Subtype of Person. Performs roles in films.                                            |
| Director           | Subtype of Person. Oversees artistic and narrative aspects of a film.                  |
| Production Company | Entity responsible for financing, production, or distribution.                         |
| Genre              | Category such as drama, comedy, action, arthouse, etc.                                 |
| Rating             | Numerical quality score (e.g., IMDB score, viewer rating).                             |
| Review Count       | Number of user reviews or views (used for detecting underrated films).                 |
| Release Year       | Year the film premiered.                                                               |
| Collaboration      | Repeated or significant cooperation between artists (useful for Persona 3).            |

### 5.1. Core Concepts

## 6. Relationships

**Table 2:** Relationships

| Relationship | From  | To                 |
|--------------|-------|--------------------|
| Acted in     | Actor | Movie              |
| Directed by  | Movie | Director           |
| Produced by  | Movie | Production Company |
| Has genre    | Movie | Genre              |
| Released in  | Movie | Release Year       |
| Has rating   | Movie | Rating             |
| Has reviews  | Movie | Review Count       |

## 7. ER Diagram (Conceptual Model)

### Entities:

- Movie: ID, Name, Release Year, Type, Genre, Rating, Runtime
- Actor: ID, Name, Birth date, Profession
- Director: ID, Name
- Production: ID, Name

### Relationships:

- Movie  $\rightarrow$  Actor (`acted_in`)
- Movie  $\rightarrow$  Director (`directed_by`)
- Movie  $\rightarrow$  Production (`produced_by`)

## 8. Focus Classification

## 9. PF-Sheet

**Table 3:** Focus Classification

| Concept               | Focus Level |
|-----------------------|-------------|
| Movie                 | High        |
| Actor                 | High        |
| Director              | High        |
| Genre                 | Medium      |
| Rating / Review Count | High        |
| Production Company    | Medium      |

| Aspect         | Definition                                                                                         |
|----------------|----------------------------------------------------------------------------------------------------|
| Domain         | Mongolian cinema (films, cast, crew, studios, genres, ratings)                                     |
| Problem        | Information is scattered, unstructured, and lacks interconnection                                  |
| Objective      | Create a connected, structured knowledge graph integrating all relevant film data                  |
| Scope          | Mongolian films with metadata: title, release year, director, cast, production company, genre      |
| Expected Users | Researchers, film enthusiasts, media organizations                                                 |
| Output         | Knowledge graph (movies, actors, directors, production; edges: acted_in, directed_by, produced_by) |

## PHASE 2 - Information Gathering

In this section, the second main input of the project is described, namely the list of data sources and datasets available for building the Knowledge Graph of Mongolian Cinema. The objective of this phase is to collect and organize the raw knowledge needed to build a unified, structured representation of Mongolian cinema data. Unlike Hollywood or Korean cinema ecosystems, which already benefit from mature, interconnected knowledge graphs that allow users to easily find release year, director, production company, genre, rating, reviews, and filmographies in one place, the Mongolian film domain lacks such a centralized knowledge infrastructure. Our goal is to gather, clean, standardize, and prepare these resources to enable the creation of such a connected ecosystem.

### 1. Sources Identification

- **Wikipedia Mongolia (mn.wikipedia.org)** Provides semi-structured pages for individual Mongolian films, including titles, directors, actors, production companies, runtimes, summaries, and award information. Type: Language + Data-value resource. Diversity Awareness: Not diversity-aware; structure varies between pages.
- **Kinosan.mn** A local Mongolian online movie portal listing films, posters, cast, genre tags, summaries, and sometimes ratings and release years. Type: Data-value resource. Diversity Awareness: Not diversity-aware; heterogeneous formats.
- **Domain Vocabulary Notes** Informal text resources describing film concepts (Movie, Person, Actor, Director, Genre, Rating). Type: Language resource. Diversity Awareness: Yes; domain vocabulary structures are fully under project control.

### 2. Datasets Collected

The following datasets were gathered and consolidated during the information gathering phase:

- **Movie Dataset** Contains metadata for Mongolian films:
  - Title, Release Year, Runtime
  - Genre

- Rating
- Actors
- Director
- Production Company
- **Actor Dataset** Includes:
  - Name
  - Last Name
  - Birthdate
- **Scraped Cast–Movie Mapping** Actor–Movie and Director–Movie relationships extracted from Wikipedia and Kinosan.mn. Used for constructing graph edges such as `acted_in` and `directed_by`.
- **Genre and Category Extraction** List of genres from scraped websites and manually curated classification rules.

### 3. Dataset Cleaning

A series of cleaning activities were required due to the heterogeneous and incomplete nature of Mongolian cinema sources.

- **Name normalization:** Actors and directors appear in multiple spellings on websites. Example: “ . ” vs “ - ”.
- **Duplicate removal:** Some films appear multiple times when scraped from different pages.
- **Missing values handling:** Several films lacked runtime, director, or review counts. Missing values were imputed as:
  - NULL values in the dataset
- **Standardization of release date formats:** From free-text (e.g., “2001 3- ”) to integer year values.

### 4. Dataset Standardisation

To support the creation of a unified Knowledge Graph schema, the datasets were standardized as follows:

- **Entity identifiers (IDs)** created for: Movies, Actors, Directors, Production Companies, and Genres.
- **Schema alignment:** All datasets aligned with a shared conceptual schema consistent with film ontologies and Schema.org patterns (`Movie`, `Person`, `Organization`, `CreativeWork`).
- **Relationship standardization:** Unified relationship labels:
  - `acted_in`
  - `directed_by`
  - `produced_by`
  - `has_genre`
  - `has_rating`
- **Attribute standardization:** Duration → integer (minutes) Release year → integer Rating → float Age rate → string

## 5. Discussion of Choices, Strong and Weak Points

### Strengths

- **Local relevance:** The data set reflects the actual Mongolian cinema landscape, allowing realistic KG use cases.
- **Rich metadata coverage:** The combined sources provide: title, actors, directors, genre, release year, production company, ratings, summaries, and awards — enabling meaningful graph queries.
- **Compatibility with existing global models:** The data set can be mapped to Schema.org and Wikidata film structures to ensure interoperability.

### Weaknesses

- **Heterogeneous formats:** Wikipedia and Kinolan pages differ significantly in structure and require extensive cleanup.
- **Incomplete data:** Many Mongolian films lack ratings, review counts, or award information.
- **Name inconsistency:** Lack of standardized transliteration or naming conventions for actors and directors.

**Conclusion** Despite limitations, the collected datasets (Wikipedia MN, Kinolan.mn) provide a sufficiently rich base to proceed to conceptual modeling, ontology design, and graph construction. The datasets have been cleaned, normalized, and standardized, making them ready for integration into the Mongolian Cinema Knowledge Graph.

## PHASE 3 — Language Definition

Phase 3 of the iTelos methodology, the Language Definition phase, establishes the formal representation language for the Mongolian Cinema Knowledge Graph. Building on the concepts and datasets from Phase 1 and Phase 2, an ontology-based approach is used to enable structured, machine-readable, and extensible knowledge representation. The goal of this phase is to define ontology classes, properties, constraints, and mappings that transform raw datasets into a coherent knowledge graph. Language resources include domain vocabularies, conceptual schemas, naming conventions, and filtered resource structures derived from Phase 2 datasets. Project-specific concepts follow the identifier convention MOV25- (MOV25-0 to MOV25-999) to ensure uniqueness. The producer formalizes extracted concepts, while the consumer identifies additional concepts needed to integrate resources and support competency questions, ensuring a consistent and reusable knowledge graph.

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### 1. Ontology Scope and Design Principles

**Domain:** Mongolian Cinema

The ontology models films, people involved in film production, production companies, genres, ratings, and their relationships.

**Scope limitations:**

- Focus on Mongolian feature films and their primary cast and crew.
- One director per movie is assumed for simplicity at this stage.
- Ratings and review counts are modeled as literal values.



### Design principles:

- Alignment with existing vocabularies such as Schema.org and Wikidata where possible.
  - Simplicity and clarity over excessive detail.
  - Extensibility to support future phases (recommendation, analytics).
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## 2. Class Hierarchy

The ontology defines the following main classes and subclass relationships:

- **CreativeWork**
  - Movie
- **Person**
  - Actor
  - Director
- **Organization**
  - ProductionCompany
- **Genre**
- **Award**

This hierarchy allows shared properties (e.g., name, identifier) to be inherited while supporting role-specific semantics.

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## 3. Object Properties

Object properties define relationships between entities in the knowledge graph.

**Table 4:** Object Properties

| Property          | Domain | Range             |
|-------------------|--------|-------------------|
| acted_in          | Actor  | Movie             |
| directed_by       | Movie  | Director          |
| produced_by       | Movie  | ProductionCompany |
| has_genre         | Movie  | Genre             |
| won_award         | Movie  | Award             |
| collaborates_with | Person | Person            |

The `collaborates_with` relationship is a derived relationship based on repeated co-appearances in films and is primarily used to support business-oriented analysis (Persona 3).

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## 4. Data Properties

Data properties describe attributes of entities using literal values.

## Movie

- title (string)
- releaseYear (integer)
- runtime (integer)
- rating (float)
- reviewCount (integer)
- summary (string)

## Person

- name (string)
- birthDate (date)

## Organization

- name (string)
- 

## 5. Modeling Constraints and Assumptions

To ensure consistency and feasibility, the following modeling decisions are applied:

- Each Movie is associated with exactly one Director.
- A Movie may be associated with multiple Actors.
- Ratings and review counts are modeled as data properties rather than entities.
- Missing values are allowed and represented as NULL.

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These constraints simplify graph construction while preserving analytical usefulness.

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## 6. Dataset-to-Ontology Mapping

The collected datasets from Phase 2 are mapped to ontology elements as follows:

**Table 5:** Dataset to Ontology Mapping

| Dataset Field      | Ontology Element                     |
|--------------------|--------------------------------------|
| Movie Title        | Movie.title                          |
| Release Year       | Movie.releaseYear                    |
| Actor Name         | Actor.name + acted_in                |
| Director Name      | Director.name + directed_by          |
| Production Company | ProductionCompany.name + produced_by |
| Genre              | Genre + has_genre                    |
| Rating             | Movie.rating                         |
| Review Count       | Movie.reviewCount                    |

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This mapping enables automated transformation of cleaned datasets into graph triples.

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## 7. Example Knowledge Graph Instances

The following example illustrates an instantiated fragment of the knowledge graph:

Movie:

```
releaseYear: 1995
has_genre: Drama
directed_by: .
acted_in: Actor_1, Actor_2
rating: 7.6
```

Such instances demonstrate how real-world film data is represented using the defined ontology.

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## 8. Readiness for Graph Construction

The defined ontology satisfies all competency questions introduced in Phase 1 and is fully aligned with the standardized datasets prepared in Phase 2.

This language definition provides a solid foundation for the next phase, where RDF triples or property graphs will be generated and stored in a graph database for querying and analysis.